

Full Stack Development with MERN

Project Documentation

1. Introduction

Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

Team Members:

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2. Project Overview

Purpose:

The project is designed to improve security in ServiceNow by controlling record access based on user roles and specific record field values. Script-controlled Access Control Lists (ACLs) are used to prevent unauthorized users from viewing or modifying sensitive records.

Features:

- Role-based record access control
- Field-value-based access restriction
- READ, CREATE, WRITE, and DELETE access control
- Script-based ACL evaluation
- Prevention of unauthorized record access
- Secure access through ServiceNow Forms, Lists, and Service Portal

3. Architecture

Frontend:

The project uses the ServiceNow Web User Interface, including Forms, Lists, and Service Portal, through which users interact with records.

Backend:

ServiceNow ACLs and server-side JavaScript scripts evaluate the user's role and the record's field values before allowing or denying access.

Database:

ServiceNow platform records are stored and managed within the ServiceNow platform database. ACL rules control access to these records based on defined conditions.

4. Setup Instructions

Prerequisites:

- ServiceNow instance
- Valid ServiceNow user account
- Appropriate administrator/developer access
- Web browser

Installation:

The project is configured directly within the ServiceNow platform. Users, roles, tables, and ACL rules are created and configured in the ServiceNow instance. Script conditions are added to the required ACLs to control record access.

5. Folder Structure

The project does not use a traditional client-server folder structure because it is developed on the ServiceNow platform. The main project components include users, roles, tables, Access Control Lists, ACL scripts, and test configurations.

6. Running the Application

The application runs directly on the ServiceNow cloud platform. Users log in to the ServiceNow instance through a web browser and access the configured records. The ACL engine automatically evaluates access permissions when users perform READ, CREATE, WRITE, or DELETE operations.

7. API Documentation

No external REST API is required for this project. Access control is implemented using ServiceNow ACL rules and server-side JavaScript. The ACL engine evaluates user roles and record field values before granting access to records.

8. Authentication

ServiceNow user authentication is used to identify users and their assigned roles. After authentication, the ACL system evaluates the user's roles and the relevant record field values.

Based on these conditions, the system allows or denies READ, CREATE, WRITE, and DELETE operations.

9. User Interface

The project uses the standard ServiceNow interface, including record Forms, Lists, and Service Portal. Users can access records according to their assigned permissions, while unauthorized users are prevented from accessing restricted records.

10. Testing

The project was tested using functional and access-control test cases. READ, CREATE, WRITE, and DELETE operations were tested for authorized and unauthorized users. Role-based and field-value-based access conditions were also verified to ensure that restricted records could not be accessed without proper authorization.

11. Screenshots or Demo

The project demonstration shows the configuration and working of the Script-Controlled ACLs in ServiceNow. The demonstration covers user roles, ACL configuration, access validation, and prevention of unauthorized record access.

12. Known Issues

The project depends on correct ServiceNow role assignments, ACL configurations, and field-value conditions. Incorrect configuration of these rules may result in users receiving or being denied access unexpectedly.

13. Future Enhancements

Future enhancements can include more advanced field-based access rules, additional user roles, automated ACL testing, improved access-monitoring features, detailed security reports, and integration with other ServiceNow applications.

Conclusion

The Script-Controlled ACL project provides a secure and flexible method for controlling ServiceNow record access. By combining user roles with record field values, the system provides controlled READ, CREATE, WRITE, and DELETE permissions and helps protect sensitive records from unauthorized access.